

Optical Networks and Systems (ONS)

Tutorial 3

Tutorial Date: November 30, 2022, 14:00

Solution to Problem 1:

We can find that,

$$\begin{aligned} I_{\text{bias}} &= I_{\text{min}} + \frac{I_{\text{max}} - I_{\text{min}}}{2} \\ &= \frac{I_{\text{max}} + I_{\text{min}}}{2} \end{aligned}$$

For PAM-2 format, we consider $s_{\text{min}} = -1$ and $s_{\text{max}} = +1$. We can that write I_{min} and I_{max} as follows,

$$\begin{aligned} I_0 s_{\text{min}} + I_{\text{bias}} &= I_{\text{min}} \\ I_0 s_{\text{max}} + I_{\text{bias}} &= I_{\text{max}} \end{aligned}$$

From the above two equations we get,

$$I_0 = \frac{I_{\text{min}} - I_{\text{bias}}}{s_{\text{min}}} = \frac{I_{\text{max}} - I_{\text{bias}}}{s_{\text{max}}} = \frac{I_{\text{max}} - I_{\text{min}}}{2}$$

For PAM-4 format, we consider $s_{\text{min}} = -3$ and $s_{\text{max}} = +3$. Similarly, we get,

$$I_0 = \frac{I_{\text{min}} - I_{\text{bias}}}{s_{\text{min}}} = \frac{I_{\text{max}} - I_{\text{bias}}}{s_{\text{max}}} = \frac{I_{\text{max}} - I_{\text{min}}}{6}$$

The average drive current in both cases is I_{bias} . Considering all symbols are equally likely, the variance of the drive current for a PAM-M is then,

$$\begin{aligned} \sigma^2 &= \frac{1}{M} \sum_{i=1}^M (I_i - I_{\text{bias}})^2 \\ &= \frac{1}{M} \sum_{i=1}^M (I_0 s_i)^2 \end{aligned}$$

For PAM-2 format we can then calculate,

$$\sigma^2 = \frac{I_0^2}{2} [(-1)^2 + (+1)^2] = (I_{\text{max}} - I_{\text{min}})^2 \frac{2}{8} = 0.25 \times (I_{\text{max}} - I_{\text{min}})^2$$

For PAM-4 format we can then calculate,

$$\sigma^2 = \frac{I_0^2}{4} [(-3)^2 + (-1)^2 + (+1)^2 + (+3)^2] = (I_{\text{max}} - I_{\text{min}})^2 \frac{20}{144} = 0.1389 \times (I_{\text{max}} - I_{\text{min}})^2$$

Reduction of signal power in dB,

$$10 \log_{10} \left(\frac{20 \times 8}{144 \times 2} \right) = -2.59 \text{ dB}$$

Solution to Problem 2:

a) Optical transceiver is an optical module consisting of both an optical transmitter and an optical receiver.

- TOSA stands for Transmit Optical Sub-Assembly:
 - Primary function is to convert an electrical signal into an optical signal.
 - Typically contains a light source (LED or Laser diode) and electrical interface.
- ROSA stands for Receiver Optical Sub-Assembly
 - Primary function is to convert an optical signal into an electrical signal.

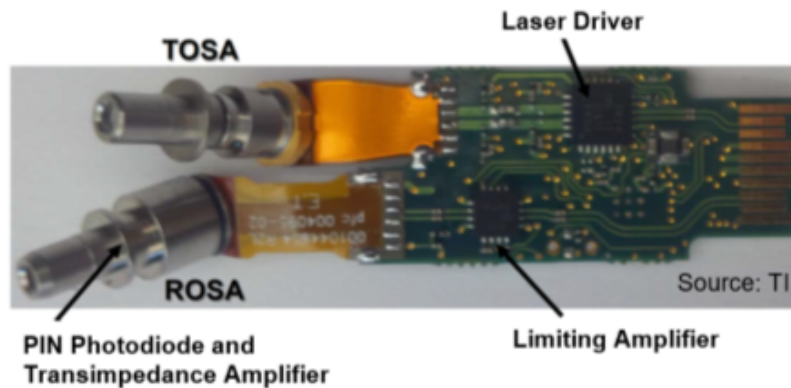


Figure 1: Optical transceiver.

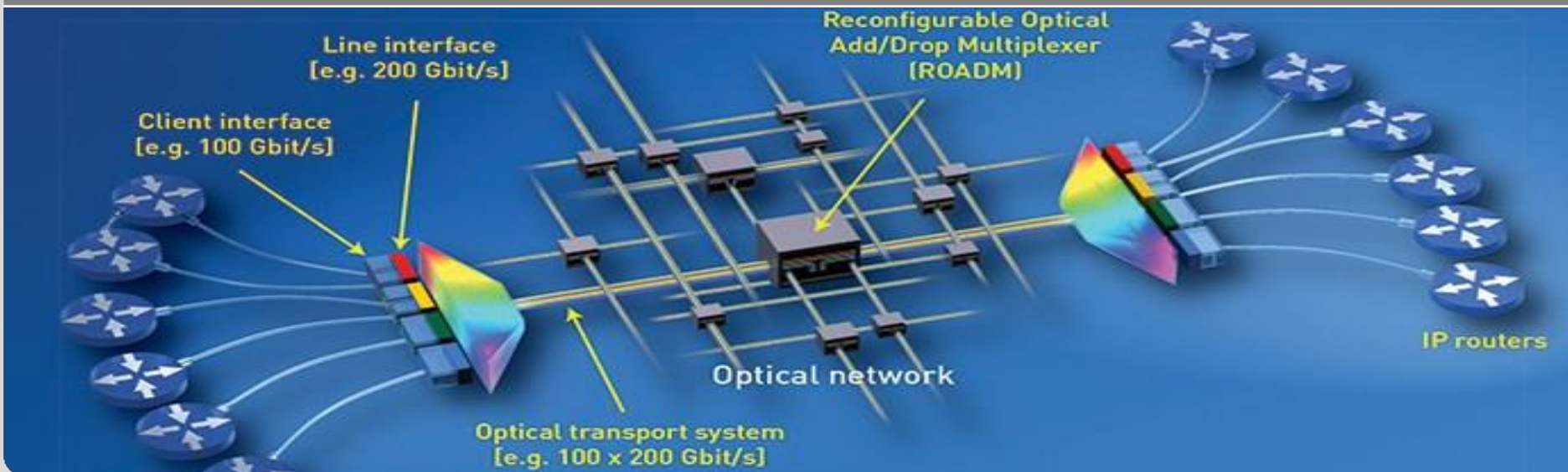
b) There are three fundamental processes found in semiconductor devices which are utilized in optical transceivers. Describe these processes and their applications.

- Absorption: Incident light is absorbed and results in an electron-hole pair. An example device is a photodetector.
- Spontaneous emission: An electron falls from an excited state to a lower energy state with no outside mechanism emitting a photon. Example device, light emitting diodes.
- Stimulated emission: An electron falls from an excited state to a lower energy state due to an incident photon whilst emitting a photon. Example device is a Laser.

ONS: Tutorial 2

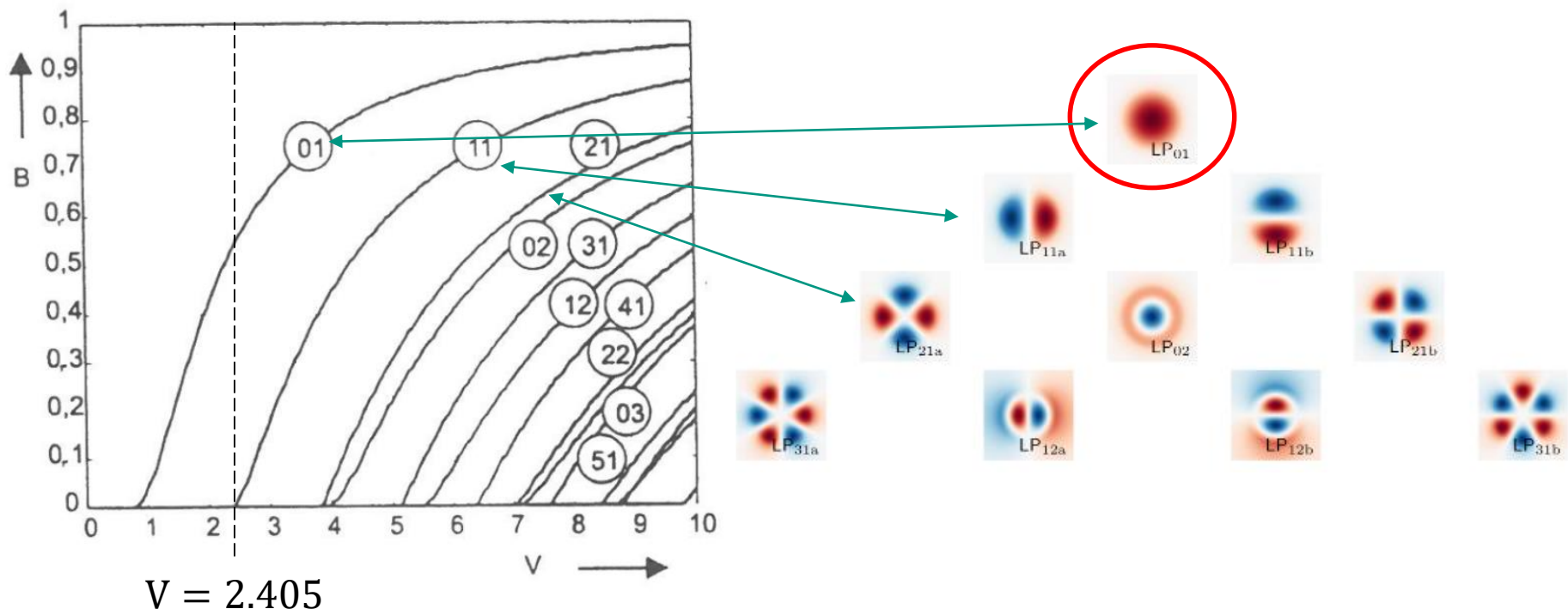
Salek Mahmud

Institute of Photonics and Quantum Electronics (IPQ), Department of Electrical Engineering and Information Technology (ETIT)



Problem 1: Optical Fiber Acceptance Angle

An optical signal is sent over an optical fiber. For a weakly guiding fiber, the transverse field distribution $F_m(x, y)$ can be characterized by linearly polarized (LP) modes. Figure 1a depicts the normalized propagation constant B of different LP-modes as a function of the normalized frequency V for a step-index fiber. As can be seen in the figure, single-mode operation is obtained for $V < 2.405$ only. Calculate the critical acceptance angles for single-mode fibers with core radii of $1\ \mu\text{m}$, $4\ \mu\text{m}$, $10\ \mu\text{m}$ and $50\ \mu\text{m}$. Assume $\lambda = 1550\ \text{nm}$.



Problem 1: Optical Fiber Acceptance Angle

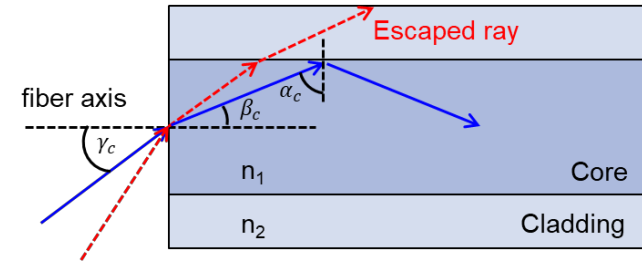
Calculate the critical acceptance angles for single-mode fibers with core radii of 1 μm , 4 μm , 10 μm , and 50 μm . Assume $\lambda = 1550 \text{ nm}$

- To Maintain single mode operation we need $V < 2.405$
- V parameter is defined as: $V = ak_0A_N$

a = fiber core radius

$$k_0 = 2\pi/\lambda$$

$$A_N = \text{numerical aperture} = \sin(\gamma_c)$$



Step I. Find values of A_N for various core radii and $V = 2.405$.

Step II. The acceptance angle in radian can be found as: $\gamma_c = \arcsin(A_N)$

Radius (a) [μm]	A_N	γ_c [Degrees]
1	0.6	36
4	0.15	8.6
10	0.06	3.4
50	0.012	0.7

Problem 2: Gaussian Pulse Shape

- In digital communication systems, a sequence of bits is modulated onto an analog waveform $s(t)$, which is then transmitted over the physical medium (e.g., a copper wire, an optical fiber or a wireless link).
- This process is typically carried out in two steps:
 - (I) The bit sequence is mapped to a sequence $\{a_k\}$ of transmit symbols a_k from a finite alphabet, e.g. $\{0,1,2,3\}$ in case of four levels.
 - (II) The discrete-time sequence of symbols with temporal spacing T_{sym} is converted into an analog waveform according to

$$s(t) = \sum_k a_k p(t - kT_{\text{sym}})$$

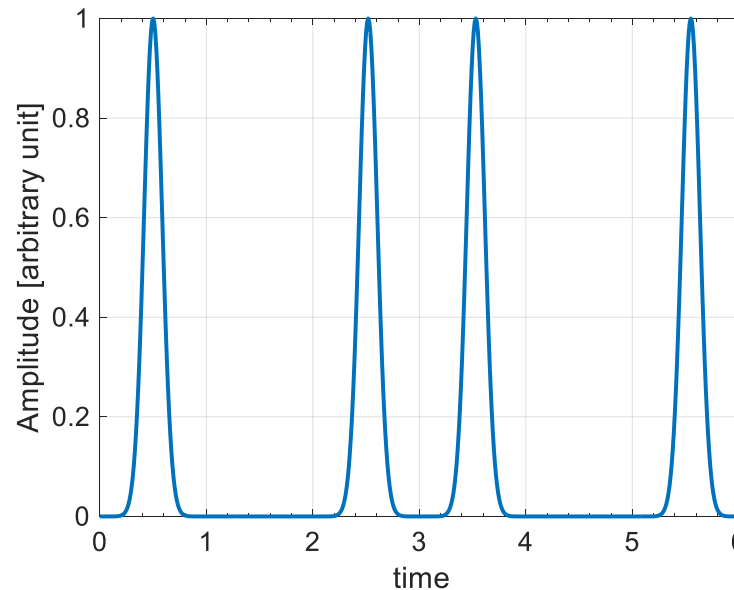
Here, $p(t)$ is called the pulse shape. One example is a Gaussian pulse with full-width-half maximum T_{fwhm} and

$$p(t) = \exp\left(-4 \ln(2) \frac{t^2}{T_{\text{fwhm}}^2}\right)$$

Problem 2: Gaussian Pulse Shape

- Plot the pulse shape and label the axes
- Compute and plot the Fourier transform of $p(t)$
- What is the two-sided bandwidth and how does it relate to the pulse width?

- Example: Consider a bit sequence $\{1, 0, 1, 1, 0, 1\}$ to be sent and gaussian pulse shape. The figure below shows the analog waveform $s(t)$ over time.



Problem 2: Gaussian Pulse Shape

- Plot the pulse shape and label the axes

- At $t = \pm \frac{T_{\text{fwhm}}}{2}$

$$\begin{aligned}
 p_{\pm \frac{T_{\text{fwhm}}}{2}} &= \exp\left(-4 \ln(2) \frac{T_{\text{fwhm}}^2}{4T_{\text{fwhm}}^2}\right) \\
 &= \exp(-\ln(2)) \\
 &= \exp\left(\ln\left(\frac{1}{2}\right)\right) \\
 &= \frac{1}{2}
 \end{aligned}$$

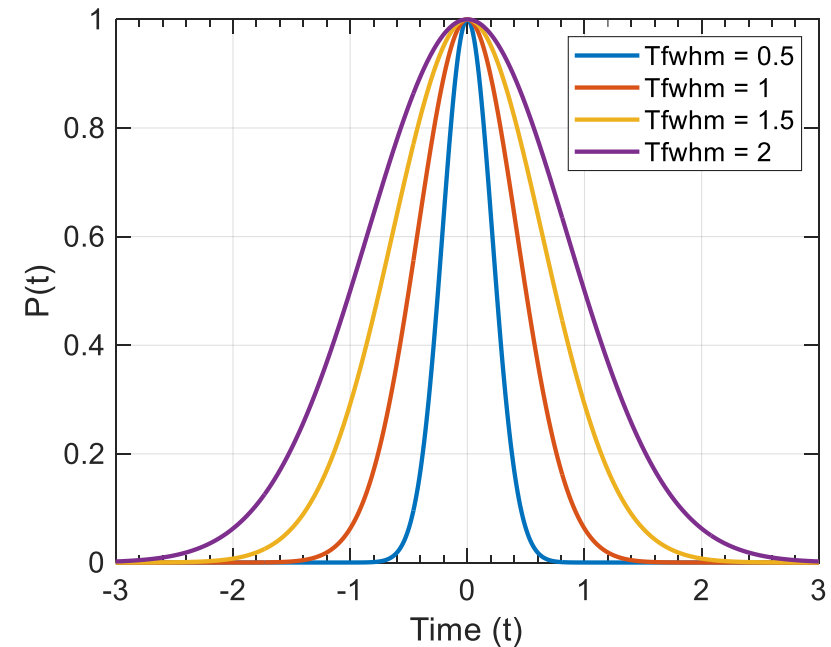
$$p(t) = \exp\left(-4 \ln(2) \frac{t^2}{T_{\text{fwhm}}^2}\right)$$

- Example plotting using Matlab

```
% generating a gaussian distribution
for various FWHM
```

```
t = -3:0.01:3;
Tfwhm = [0.5 1 1.5 2];
pt = zeros(length(Tfwhm),length(t));

for n = 1 : length(Tfwhm)
    pt(n, :) = exp(-1*4*log(2)/Tfwhm(n)^2.*(t.^2));
    plot(t,pt(n,:))
    hold on
end
```



Problem 2: Gaussian Pulse Shape

- Compute and plot the Fourier transform of $p(t)$,

$$p(t) = \exp\left(-4 \ln(2) \frac{t^2}{T_{\text{fwhm}}^2}\right)$$



$$P(f) = \sqrt{\frac{\pi}{\frac{4 \ln 2}{T_{\text{fwhm}}^2}}} \exp\left(-\frac{\pi^2 f^2}{\frac{4 \ln 2}{T_{\text{fwhm}}^2}}\right).$$

From Fourier transform table

$$x(t) = \exp(-at^2)$$



$$X(f) = \sqrt{\frac{\pi}{a}} \exp\left(-\frac{\pi^2 f^2}{a}\right)$$

- So the Fourier Transform of a Gaussian pulse is also a Gaussian

$$P(f) = \sqrt{\frac{\pi}{\frac{4 \ln 2}{T_{\text{fwhm}}^2}}} \exp\left(-\frac{4 \ln(2) f^2}{\frac{(4 \ln 2)^2}{\pi^2 T_{\text{fwhm}}^2}}\right)$$

- Substituting two-sided bandwidth $F_{\text{fwhm}} = \frac{4 \ln(2)}{\pi T_{\text{fwhm}}}$ gives

$$P(f) = \sqrt{\frac{\pi}{\frac{4 \ln 2}{T_{\text{fwhm}}^2}}} \exp\left(-\frac{4 \ln(2) f^2}{F_{\text{fwhm}}^2}\right)$$

Problem 2: Gaussian Pulse Shape

- What is the two-sided bandwidth and how does it relate to the pulse width?

$$p(t) = \exp\left(-4 \ln(2) \frac{t^2}{T_{\text{fwhm}}^2}\right)$$

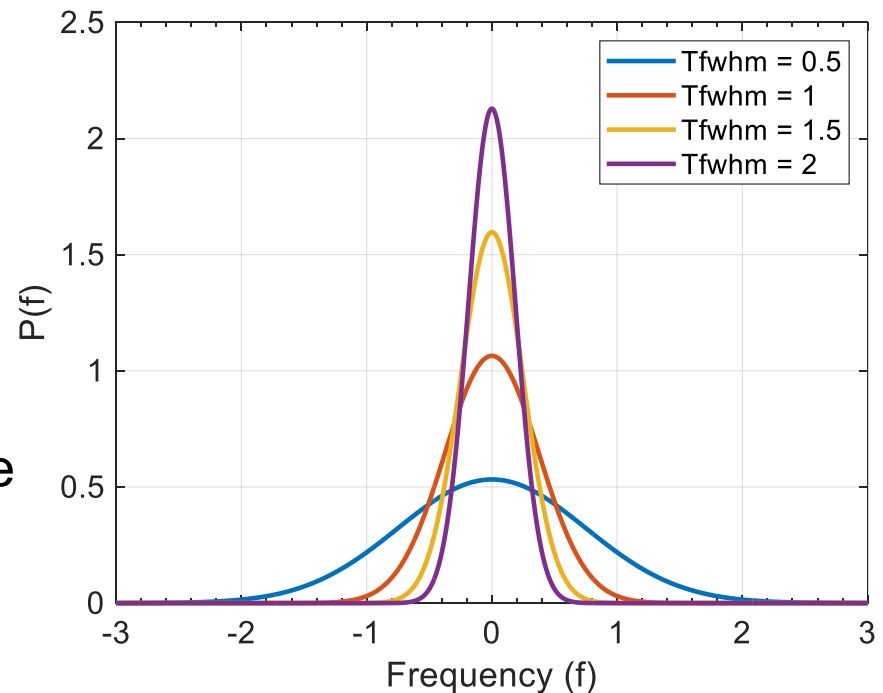


$$P(f) = \sqrt{\frac{\pi}{\frac{4 \ln 2}{T_{\text{fwhm}}^2}}} \exp\left(-\frac{\pi^2 f^2}{\frac{4 \ln 2}{T_{\text{fwhm}}^2}}\right)$$

- Two-sided bandwidth relates to the pulse width as follows:

$$F_{\text{fwhm}} = \frac{4 \ln(2)}{\pi T_{\text{fwhm}}}$$

$$F_{\text{fwhm}} \times T_{\text{fwhm}} = 0.8825$$



Optical Networks and Systems (ONS)

Tutorial 7: Solution

Tutorial Date: January 18, 2023, 14:00

Solution to Problem 1: Coherent optical receivers

- a) Adding the additional complex field amplitudes $\underline{A}_1$ to $\underline{A}_{12}$ in Figure 1 allows us to separate the 90° hybrid into four separate 180° hybrids and treat the 90° hybrid in three stages. Using the formula for the 180° hybrid we find for the field amplitudes $\underline{A}_1$ to $\underline{A}_{12}$:

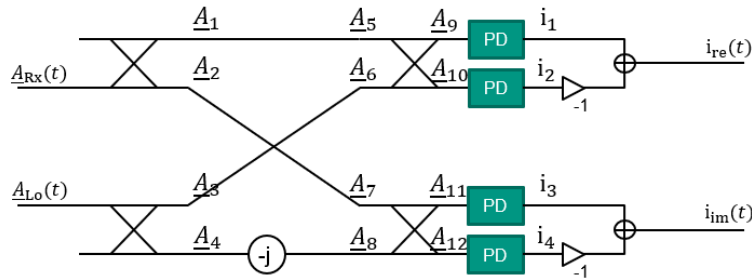


Figure 1

$$\begin{pmatrix} \underline{A}_1 \\ \underline{A}_2 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & j \\ j & 1 \end{pmatrix} \begin{pmatrix} 0 \\ \underline{A}_{Rx} \end{pmatrix} \quad (1)$$

$$\begin{pmatrix} \underline{A}_3 \\ \underline{A}_4 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & j \\ j & 1 \end{pmatrix} \begin{pmatrix} \underline{A}_{Lo} \\ 0 \end{pmatrix} \quad (2)$$

$$\underline{A}_5 = \underline{A}_1 \quad (3)$$

$$\underline{A}_6 = \underline{A}_3 \quad (4)$$

$$\underline{A}_7 = \underline{A}_2 \quad (5)$$

$$\underline{A}_8 = -j\underline{A}_4 \quad (6)$$

$$(7)$$

$$\begin{pmatrix} \underline{A}_9 \\ \underline{A}_{10} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & j \\ j & 1 \end{pmatrix} \begin{pmatrix} \underline{A}_5 \\ \underline{A}_6 \end{pmatrix} \quad (8)$$

$$\begin{pmatrix} \underline{A}_{11} \\ \underline{A}_{12} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & j \\ j & 1 \end{pmatrix} \begin{pmatrix} \underline{A}_7 \\ \underline{A}_8 \end{pmatrix} \quad (9)$$

Combining equations 1 to 9 allows us to express the complex field amplitudes in front of the photodetectors in terms of $\underline{A}_{Rx}$ and $\underline{A}_{Lo}$:

$$\begin{pmatrix} \underline{A}_9 \\ \underline{A}_{10} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} j\underline{A}_{Rx} + j\underline{A}_{Lo} \\ -\underline{A}_{Rx} + \underline{A}_{Lo} \end{pmatrix} \quad (10)$$

$$\begin{pmatrix} \underline{A}_{11} \\ \underline{A}_{12} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \underline{A}_{Rx} + j\underline{A}_{Lo} \\ j\underline{A}_{Rx} + \underline{A}_{Lo} \end{pmatrix} \quad (11)$$

The photocurrent generated in the first photodiode can be expressed as:

$$\begin{aligned} i_1 &= R_D |\underline{A}_9|^2 \\ &= R_D |j\underline{A}_{Rx} + j\underline{A}_{Lo}|^2 \\ &= R_D (j\underline{A}_{Rx} + j\underline{A}_{Lo})(j\underline{A}_{Rx} + j\underline{A}_{Lo})^* \\ &= R_D (\Re \{ \underline{A}_{Rx} \}^2 + \Im \{ \underline{A}_{Rx} \}^2 + 2 (\Re \{ \underline{A}_{Rx} \} \Re \{ \underline{A}_{Lo} \} + \Im \{ \underline{A}_{Rx} \} \Im \{ \underline{A}_{Lo} \}) \\ &\quad + \Re \{ \underline{A}_{Lo} \}^2 + \Im \{ \underline{A}_{Lo} \}^2) \\ &= \frac{R_D}{4} (|\underline{A}_{Rx}|^2 + 2 (\Re \{ \underline{A}_{Rx} \underline{A}_{Lo}^* \}) + |\underline{A}_{Lo}|^2) \end{aligned}$$

The same treatment for the currents produced by the other three photodiodes yields:

$$i_2 = R_D |\underline{A}_{10}|^2 = \frac{R_D}{4} (|\underline{A}_{Rx}|^2 - 2 (\Re \{ \underline{A}_{Rx} \underline{A}_{Lo}^* \}) + |\underline{A}_{Lo}|^2) \quad (12)$$

$$i_3 = R_D |\underline{A}_{11}|^2 = \frac{R_D}{4} (|\underline{A}_{Rx}|^2 + 2 (\Im \{ \underline{A}_{Rx} \underline{A}_{Lo}^* \}) + |\underline{A}_{Lo}|^2) \quad (13)$$

$$i_4 = R_D |\underline{A}_{12}|^2 = \frac{R_D}{4} (|\underline{A}_{Rx}|^2 - 2 (\Im \{ \underline{A}_{Rx} \underline{A}_{Lo}^* \}) + |\underline{A}_{Lo}|^2) \quad (14)$$

For the combined currents $i_{re}(t)$ and $i_{im}(t)$, we then find:

$$i_{re}(t) = i_1 - i_2 = R_D \Re \{ \underline{A}_{Rx}(t) \underline{A}_{Lo}^*(t) \} \quad (15)$$

$$i_{im}(t) = i_3 - i_4 = R_D \Im \{ \underline{A}_{Rx}(t) \underline{A}_{Lo}^*(t) \} \quad (16)$$

- b) QPSK transmits 2 bit per symbol. Thus the symbol rate is half the data rate: $R_{sym} = \frac{1}{2} R_{data}$. The phase rotation for the n-th received symbol can be found as:

$$\begin{aligned} \Delta\varphi(t) &= \arg \left(R_D \underline{A}_{Rx}(t) |\underline{A}_{Lo}| e^{-j2\pi\Delta f_{Lo}t} e^{-j\Delta\varphi_{Lo}(t)} \right) \\ &= -2\pi\Delta f_{Lo}t - \Delta\varphi_{Lo}(t) \end{aligned}$$

Taking into consideration, that we assumed the phase noise to be zero and using the definition of the frequency offset from the lecture with $\Delta f_{Lo} = f_{Lo} - f_{Ref} = -3.125$ GHz, we find the phase rotation to be:

$$\Delta\varphi(t_n) = 2\pi 3.125 \text{ GHz} (n-1) T_{sym}$$

Where $T_{sym} = \frac{1}{R_{sym}}$ is the symbol period.

For the first, second and n-th symbol we can calculate the phase rotation as:

$$\begin{aligned} \Delta\varphi(t_1 = 0) &= 2\pi 3.125 \text{ GHz} 0 \frac{1}{R_{sym}} \\ \Delta\varphi(t_2 = T_{sym}) &= 2\pi 3.125 \text{ GHz} \frac{1}{25 \text{ GBaud}} = 2\pi \frac{1}{8} = 45^\circ \\ \Delta\varphi(t_n = (n-1)T_{sym}) &= 2\pi (n-1) \frac{1}{8} \end{aligned}$$

Optical Networks and Systems (ONS)

Tutorial 6: Solution

Tutorial Date: January 11, 2023, 14:00

Solution to Problem 1: Cyclic redundancy check

a.) To find the CRC, the message is padded with four zeros, equal to the length of the resulting CRC. The resulting padded message is divided by the generator sequence using XOR starting at the most significant bit. The resulting rest is the CRC.

$$\begin{array}{r}
 \overline{1101100000} \\
 110101 \quad \leftarrow g(x) \\
 \hline
 0000110000 \\
 \quad 110101 \\
 \hline
 \quad \quad 000101
 \end{array}$$

$\Rightarrow$ 11011 00101

Figure 1

b.) To check the arrived messages for errors, the arrived message is divided by the generator sequence via XOR. If the rest is unequal to zero, the received message contained a bit error.

$ \begin{array}{r} 1101100101 \\ 110101 \\ \hline 0000110101 \\ \quad 110101 \\ \hline \quad \quad 000000 \end{array} $ $\Rightarrow$ No Error	$ \begin{array}{r} 1001100101 \\ 110101 \\ \hline 0100110 \\ \quad 110101 \\ \hline \quad \quad 0100111 \\ \quad \quad \quad 110101 \\ \hline \quad \quad \quad 0100100 \\ \quad \quad \quad \quad 110101 \\ \hline \quad \quad \quad \quad 0100011 \\ \quad \quad \quad \quad \quad 110101 \\ \hline \quad \quad \quad \quad \quad 010110 \end{array} $ $\Rightarrow$ Error
--	---

Figure 2

Solution to Problem 2: System Cost Comparison

Using the transceiver costs and the fiber costs per km, the length dependent cost for each of the four possible systems can be derived:

$$\begin{aligned}C_{\text{SR4}}(L) &= 4 \times 1.6 \times L + 2 \\C_{\text{PSM4}}(L) &= 4 \times 0.1 \times L + 3 \\C_{\text{SWDM4}}(L) &= 1 \times 1.6 \times L + 4 \\C_{\text{CWDM4}}(L) &= 1 \times 0.1 \times L + 6\end{aligned}$$

The evolution of the cost for each system over the fiber length is depicted in Figure 3:

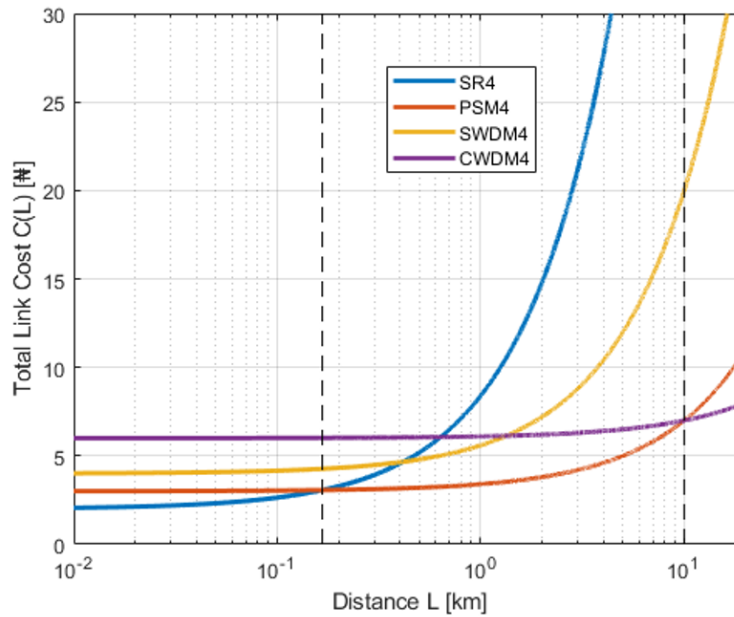


Figure 3: Evolution of the cost of the system over the fiber length.

For the most cost effective system follows:

- For $L < 167m$: SR4
- For $167m \leq L \leq 10km$: PSM4
- For $L > 10km$: CWDM4
- SWDM4 is not sensible for any length L

Solution to Problem 3: PON Design

- a) In a time division multiplexing (TDM-PON) network, different subscribers are allocated different transmission time slots. In a wavelength division multiplexing (WDM) scheme, different subscribers (or group of subscribers) are allocated different wavelengths which can run simultaneously in the same fiber.
- b) In a TDM PON network, bursts are received at different power levels due to loss variations from different optical network units (ONUs i.e. subscribers) to the optical line termination (OLT i.e. provider). At the OLT side, a so called burst-mode receiver is required that can accommodate to the various received signal power levels, for e.g., adjusting the decision threshold accordingly. In addition fast clock synchronization is needed,
- c) The fiber losses are $a_{f,dB} = 0.25 \text{ dB/km} \times 5 \text{ dB}$. With a launch power of $P_{in,dBm} = 0 \text{ dBm}$ and a receive power of $P_{out,dBm} = -21 \text{ dBm}$ it follows for the power in the system:

$$P_{out,dBm} = P_{in,dBm} - a_{f,dB} - a_{s,dB} \quad (1)$$

Rearranging the equation for the splitter losses $a_{s,dB}$ yields:

$$a_{s,dB} = P_{in,dBm} - a_{f,dB} - P_{out,dBm} = 16 \text{ dB} \quad (2)$$

A splitter can be considered as a concatenated cascade of 1 by 2 splitters. Each 1 by 2 splitter has an inherent splitting loss of 3 dB. Additionally each splitter stage features an excess loss of 0.2 dB. It follows for the amount of splitter stages:

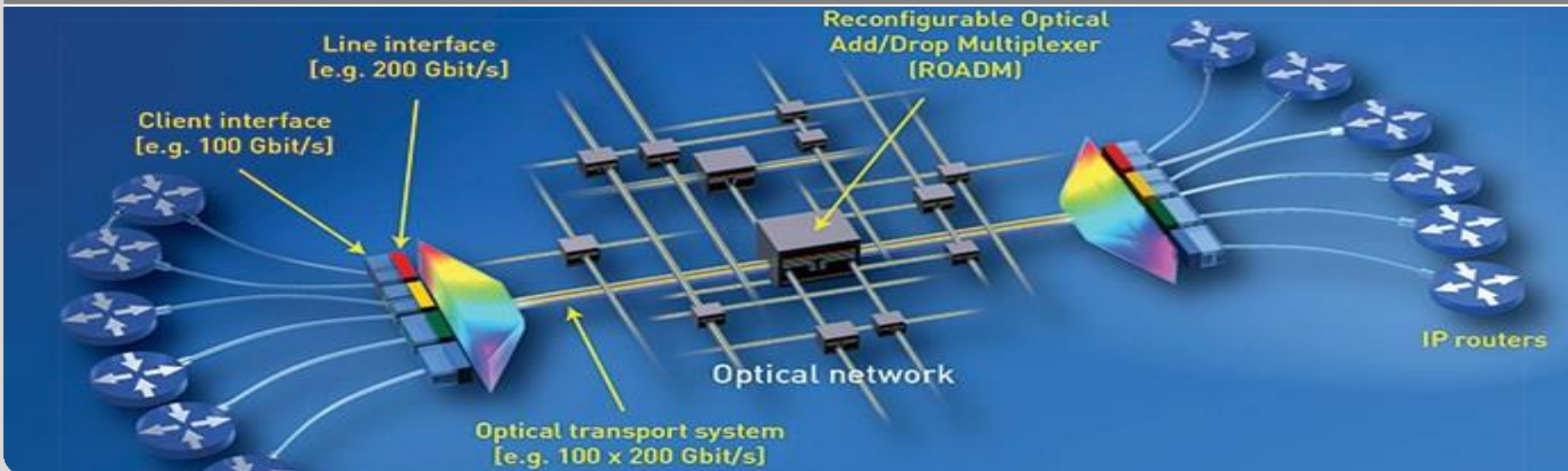
$$N = a_{s,dB} / 3.2 \text{ dB} = 5 \quad (3)$$

Each splitter stage doubles the amount of possible subscribers. Thus $2^5 = 32$ users can be serviced in the PON.

ONS – Tutorial 9 Solution

Alban Sherifaj

Institute of Photonics and Quantum Electronics (IPQ), Department of Electrical Engineering and Information Technology (ETIT)



Problem 1: Channel capacity

- a) Assume that you have a source that transmits from a binary set (source alphabet) $\{0; 1\}$, for which the probabilities are given as $p(0) = 1-p(1)$. Calculate the entropy H of this binary set. Calculate the maximum of H ; what values do $p(0)$ and $p(1)$ take in that case?

- Entropy H of this binary state is calculated as:

$$H = -\sum_i p(i) \cdot \log_2(p(i)), \text{ where } i \text{ stands for each possible value of the binary state.}$$

- Therefore, H becomes

$$\begin{aligned} H &= -p(0) \cdot \log_2(p(0)) - p(1) \cdot \log_2(p(1)) \\ &= -p(0) \cdot \log_2(p(0)) - (1-p(0)) \cdot \log_2(1-p(0)) \end{aligned}$$

- For finding the maximum of H , we take the derivative. i.e. $\partial H / \partial p(0) \triangleq 0$. For simpler evaluation of the derivative, we employ the logarithmic relation: $\log_2(p(0)) = \ln(p(0)) / \ln(2)$

$$\therefore H = \frac{1}{\ln(2)} \cdot [-p(0) \cdot \ln(p(0)) - (1-p(0)) \cdot \ln(1-p(0))]$$

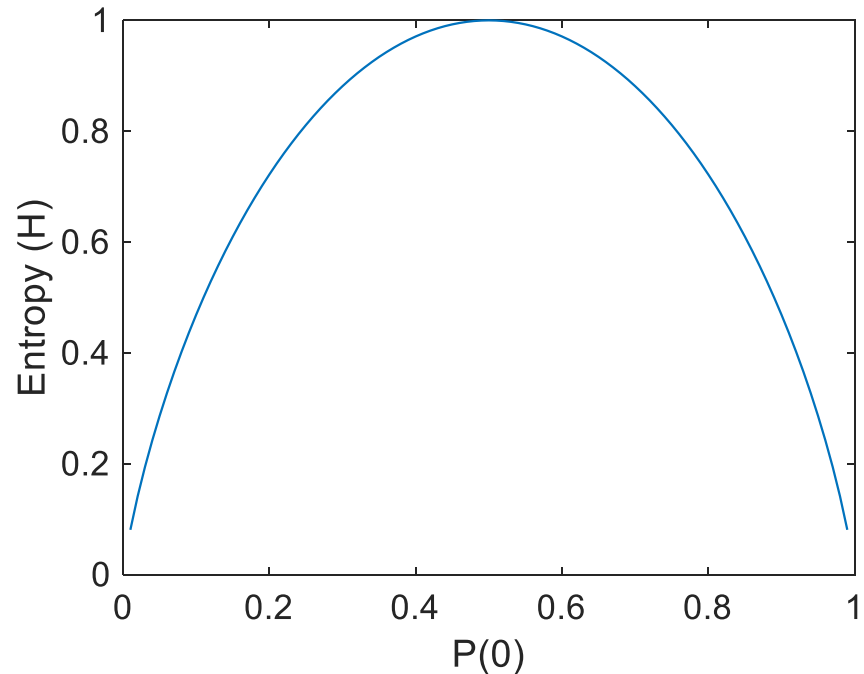
$$\frac{\partial H}{\partial p(0)} = \frac{1}{\ln(2)} \cdot \left[-\ln(p(0)) - p(0) \cdot \frac{1}{p(0)} + \ln(1-p(0)) + (1-p(0)) \cdot \frac{1}{(1-p(0))} \right] = 0$$

$$= \frac{1}{\ln(2)} \cdot [-\ln(p(0)) - 1 + \ln(1-p(0)) + 1] = 0$$

Problem 1: Channel capacity

$$\frac{\partial H}{\partial p(0)} = \frac{1}{\ln(2)} \cdot [-\ln(p(0)) + \ln(1 - p(0))] = 0$$

$$\begin{aligned} \rightarrow & -\ln(p(0)) + \ln(1 - p(0)) = 0 \\ & \ln(1 - p(0)) = \ln(p(0)) \\ & 1 - p(0) = p(0) \\ & 1 = 2p(0) \\ \Rightarrow & p(0) = 1/2 \text{ and } p(1) = 1 - p(0) = 1/2 \end{aligned}$$



- For $p(0) = 1/2$, H becomes:

$$\begin{aligned} H &= -\frac{1}{2} \cdot \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \cdot \log_2\left(\frac{1}{2}\right) \\ &= -\frac{1}{2} \cdot [\log_2\left(\frac{1}{2}\right) + \log_2\left(\frac{1}{2}\right)] \\ &= \frac{1}{2} \cdot [\log_2(2) + \log_2(2)] = 1 \text{ [bit]} \end{aligned}$$

Problem 1: Channel capacity

b) A source transmits symbols from the alphabet $\{A; B; C; D\}$. Each symbol of this source alphabet is represented accordingly using the code word $\{0; 10; 110; 111\}$ and the corresponding probabilities are

i) $\{0.25; 0.25; 0.25; 0.25\}$

ii) $\{8/15; 4/15; 2/15; 1/15\}$

Calculate the entropy of the source for these cases and compare it to the mean code word length.

i.

$$- H = -[0.25 \cdot \log_2(0.25) + 0.25 \cdot \log_2(0.25) + 0.25 \cdot \log_2(0.25) + 0.25 \cdot \log_2(0.25)]$$

$$H = -4 \cdot 0.25 \cdot \log_2(0.25)$$

$$H = +4 \cdot 0.25 \cdot \log_2(4) = 2 \text{ [bit]}$$

- Mean word length L_m is given as the sum (over all values of the alphabet) of the product of the probability of a value of the alphabet $p(i)$ and its length $L(i)$. i.e.

$$L_m = \sum_i p(i) \cdot L(i)$$

$$- L_m = [0.25 \cdot 1 + 0.25 \cdot 2 + 0.25 \cdot 3 + 0.25 \cdot 3]$$

$$- L_m = 2.25 \text{ [bit]}$$

Problem 1: Channel capacity

ii.

$$- \quad H = -\frac{1}{15} \cdot \left[8 \cdot \log_2 \left(\frac{8}{15} \right) + 4 \cdot \log_2 \left(\frac{4}{15} \right) + 2 \cdot \log_2 \left(\frac{2}{15} \right) + 1 \cdot \log_2 \left(\frac{1}{15} \right) \right] = 1.64 \text{ [bit]}$$

$$- \quad L_m = \frac{1}{15} \cdot [8 \cdot 1 + 4 \cdot 2 + 2 \cdot 3 + 1 \cdot 3] = 1.67 \text{ [bit]}$$

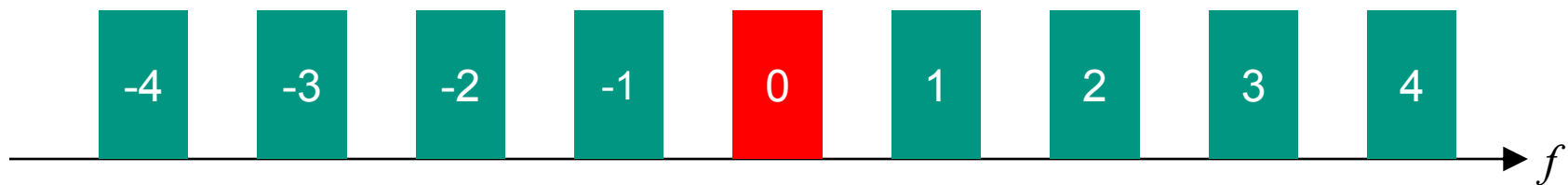
Extras: $L_m - H$ is considered as the redundancy of a code.

“Aoccdrnig to rsearch at Cmabrigde Uinervtisy, it deosn't mttar in waht oredr the ltteers in a wrod are, the olny iprmoetnt tihng is taht the frist and lsat ltteer be at the rghit pclae. The rset can be a toatl mses and you can sitll raed it wouthit a porbelm. Tihs is bcuseae the huamn mnid deos not raed ervey lteter by istlef, but the wrod as a wlohe.”

- Language (text) can be understood even if it contains errors
 - Redundancy in language
 - Not a memory-less system

Problem 1: Channel capacity

- c) In a practical system, nonlinear interference limits the capacity of optical networks after a certain launch power. Considering a WDM system, describe in words the three major nonlinear effects associated with the Kerr nonlinearity.
- Self-phase modulation (SPM): Intensity of a channel affects the phase of the signal itself. e.g. channel 0's intensity affects channel 0's phase
 - Cross-phase modulation (XPM): Intensity of neighboring channels affects the phase of a channel. e.g. channel 1's intensity affects channel 0's phase.
 - Four-wave mixing (FWM): Interaction of three independent channels affects a different fourth channel. e.g. channels 1,2, and -1 interact and affect the signal of channel 0.



Optical Networks and Systems (ONS)

Tutorial 8: Solution

Tutorial Date: February 01, 2023, 14:00

Solution to Problem 1: Concatenation of amplifiers

To solve this question, we will make use of Friis formula which is used to calculate the equivalent noise figure of a cascade of elements. For example, given a cascade of 3 optical devices with gains G_1 , G_2 and G_3 and noise figures F_1 , F_2 and F_3 as shown in Figure 2, we can summarize the first two stages as follows:

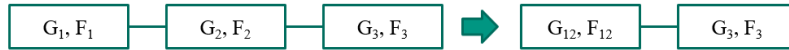


Figure 1

$$F_{12} = F_1 + \frac{F_2 - 1}{G_1} \quad (1)$$

$$G_{12} = G_1 G_2 \quad (2)$$

In the same way, we can replace the compound stage and the third stage with one single stage, combining all three stages:

$$F_{123} = F_{12} + \frac{F_3 - 1}{G_{12}} \quad (3)$$

$$G_{123} = G_{12} G_3 \quad (4)$$

This principal can be applied to an arbitrary amount of stages. Considering only one of the fiber spans, we see, that each span already consists of two stages, the fiber and the amplifier.

The losses $\mathfrak{L}$ introduced by the fiber can be described by the ratio of the input power to the output power: $\mathfrak{L} = \frac{P_{in}}{P_{out}}$. The gain of the fiber is thus defined as $G_f = \frac{1}{\mathfrak{L}}$. Since the EDFA compensates the fiber losses, its gain has to be: $G_A = \frac{1}{G_f}$. It follows, that the gain of the span is then: $G_{span} = G_A G_f = 1$. The noise figure of the fiber is $F_f = \mathfrak{L}$. With the noise figure of the amplifier F_A , we find the noise figure of the span:

$$F_{span} = F_f + \frac{F_A - 1}{G_f} = \frac{1}{G_f} + \frac{F_A - 1}{G_f} = \frac{F_A}{G_f} = \mathfrak{L} F_A \quad (5)$$

using Friis formula.

Now using formula 3 for the three concatenated stages, with the expression for each fiber span as derived above, we can express the concatenated spans as:

$$F_{123} = \mathfrak{L}_1 F_A + \frac{\mathfrak{L}_2 F_A - 1}{G_1} + \frac{\mathfrak{L}_3 F_A - 1}{G_1 G_2} \quad (6)$$

Where $\mathfrak{L}_1$, $\mathfrak{L}_2$ and $\mathfrak{L}_3$ are the length of each of the fiber spans and G_1 , G_2 and G_3 are the amplifications of the three spans. Since $G_1 = G_2 = G_3 = 1$, it follows that:

$$F_{123} = F_A (\mathfrak{L}_1 + \mathfrak{L}_2 + \mathfrak{L}_3) - 2 \approx F_A (\mathfrak{L}_1 + \mathfrak{L}_2 + \mathfrak{L}_3) \quad (7)$$

For n fiber spans, we find the general expression for the noise figure to be:

$$F = F_A \sum_{i=1}^n \mathcal{L}_i \quad (8)$$

For a fiber loss of 0.2 dB/km, applying this formula for the examples given in the task, we can calculate the noise figure F_T of the link:

		S-1	S-2	S-3	S-4	$\sum \mathcal{L}$	$F \sum \mathcal{L}$	$F_T = 10 \log_{10}(F \sum \mathcal{L})$
a	L (km)	100	100	100		300	1800	32.55
	$A = 0.2 \times L$	20	20	20				
	$\mathcal{L} = 10^{A/10}$	100	100	100				
b	L	140	80	80		710.58	4263.5	36.29
	$A = 0.2 \times L$	28	16	16				
	$\mathcal{L} = 10^{A/10}$	630.95	39.81	39.81				
c	L	80	80	140		710.58	4263.5	36.29
	$A = 0.2 \times L$	16	16	28				
	$\mathcal{L} = 10^{A/10}$	39.81	39.81	630.95				
d	L	75	75	75	75	126.49	758.95	25.80
	$A = 0.2 \times L$	15	15	15	15			
	$\mathcal{L} = 10^{A/10}$	31.62	31.62	31.62	31.62			

Solution to Problem 2: OSNR and achievable reach

a) Using the equation provided in the lecture:

$$OSNR_{dB} = 58 + P_{in,dBm} - F_{amp} - L_{span}\alpha - 10 \log_{10} N$$

As well as the values given in the task, we can solve for the equation for N :

$$10 \log_{10} N = 37 \text{ dB} - L_{span}\alpha$$

$$N = 10^{\frac{37 \text{ dB} - L_{span} 0.2 \text{ dB/km}}{10}} \quad (9)$$

Thus, we find:

L_{span}	N	L_{link}
150 km	5	750 km
100 km	50	5000 km
50 km	501	25 050 km

b) We use the equation (8) for the noise figure F of an optical link:

$$F = F_A \sum_{i=1}^n \mathcal{L}_i$$

Assuming, that each fiber link features the same length and thus the same loss, we can simplify the equation:

$$F = F_A N \mathfrak{L}$$

$$F = F_A \frac{L_{\text{link}}}{L_{\text{span}}} 10^{\frac{\alpha L_{\text{span}}}{10}}$$

In order to find the optimal span length, we have to minimize the noise figure. To do so we calculate the derivative:

$$\begin{aligned} \frac{dF}{dL_{\text{span}}} &= \frac{d}{dL_{\text{span}}} \left(F_A \frac{L_{\text{link}}}{L_{\text{span}}} 10^{\frac{\alpha L_{\text{span}}}{10}} \right) \\ &= -F_A \frac{L_{\text{link}}}{L_{\text{span}}^2} 10^{\frac{\alpha L_{\text{span}}}{10}} + F_A \frac{L_{\text{link}}}{L_{\text{span}}} \frac{d}{dL_{\text{span}}} \left(10^{\frac{\alpha L_{\text{span}}}{10}} \right) \end{aligned}$$

For the derivative of a function of the form $f(x) = a^{u(x)}$, the rule applies:

$$\frac{df}{dx} = \frac{d}{dx} a^{u(x)} = \ln(a) a^{u(x)} \frac{du(x)}{dx}$$

Using this relation, we find:

$$\begin{aligned} \frac{dF}{dL_{\text{span}}} &= -F_A \frac{L_{\text{link}}}{L_{\text{span}}^2} 10^{\frac{\alpha L_{\text{span}}}{10}} + F_A \frac{L_{\text{link}}}{L_{\text{span}}} \ln(10) 10^{\frac{\alpha L_{\text{span}}}{10}} \frac{\alpha}{10} \\ &= F_A \frac{L_{\text{link}}}{L_{\text{span}}} 10^{\frac{\alpha L_{\text{span}}}{10}} \left(\ln(10) \frac{\alpha}{10} - \frac{1}{L_{\text{span}}} \right) \end{aligned}$$

To find the minimum of the noise figure, we set the derivative to zero:

$$F_A \frac{L_{\text{link}}}{L_{\text{span}}} 10^{\frac{\alpha L_{\text{span}}}{10}} \left(\ln(10) \frac{\alpha}{10} - \frac{1}{L_{\text{span}}} \right) = 0$$

It follows:

$$\ln(10) \frac{\alpha}{10} - \frac{1}{L_{\text{span}}} = 0$$

$$\begin{aligned} L_{\text{span, opt}} &= \frac{10}{\ln(10)\alpha} \\ &= 21.71 \text{ km} \end{aligned}$$

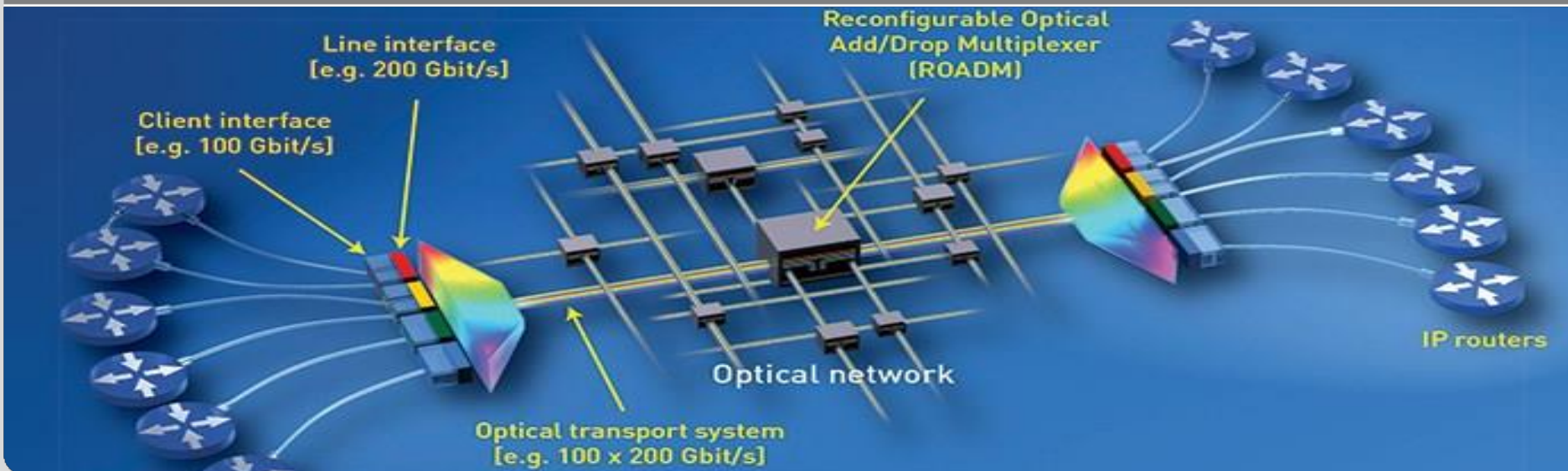
For an fiber attenuation of $\alpha = 0.2 \text{ dB/km}$. The optimum span length thus only depends on the losses in the fiber. Using formula 9, we can calculate the resulting link length:

$$\begin{aligned} N &= 10^{\frac{37 \text{ dB} - L_{\text{span, opt}} 0.2 \text{ dB/km}}{10}} = 1844 \\ \Rightarrow L_{\text{Link}} &= N \times L_{\text{span, opt}} = 40\,037 \text{ km} \end{aligned}$$

ONS: Tutorial 5

Salek Mahmud

Institute of Photonics and Quantum Electronics (IPQ), Department of Electrical Engineering and Information Technology (ETIT)



Problem 1: Dispersion

Consider a 10 Gbit/s signal is transmitted through a 625 m long multimode fiber that has a bandwidth-length product of 5 GHz · km and a chromatic dispersion of 20 ps/(km · nm). If a laser emitting at the wavelength of 1.55 μm is used, what is the total pulse broadening due to dispersion? How much is the total broadening if a 100 Gbit/s signal is transmitted through the same fiber?

■ We are given,

- Spectral width of the signal, $\Delta f = 10$ GHz
- Fiber length, $L = 0.625$ km

Note: Assuming NRZ pulse a more accurate value would be 70% of the symbol rate, $\Delta f = 7$ GHz

■ **Pulse broadening due to modal dispersion:** The effective differential group delay Δt_{MD} (denoted as, $\Delta \tau_{\text{eff}}$ in the lecture) can be written as,

$$\begin{aligned}\Delta t_{\text{MD}} &= \frac{L}{\eta_{\text{eff}}}, \\ &= \frac{0.625 \text{ km}}{5 \text{ GHz} \cdot \text{km}} \\ &= 125 \text{ ps}\end{aligned}$$

η_{eff} is the Bandwidth-length product

Problem 1: Dispersion

■ Pulse broadening due to chromatic dispersion:

$$\Delta t_{CD} = 20 \text{ ps}/(\text{km} \cdot \text{nm}) \cdot \Delta\lambda \cdot L, \quad \Delta\lambda \text{ is spectral width in nm}$$

■ Spectral width in terms of wavelength can be found as,

$$\begin{aligned}
 \Delta\lambda &= \lambda_2 - \lambda_1 \\
 &= \frac{c}{f_2} - \frac{c}{f_1} \\
 &= \frac{c \left(f_0 + \frac{\Delta f}{2} \right) - c \left(f_0 - \frac{\Delta f}{2} \right)}{\left(f_0 - \frac{\Delta f}{2} \right) \cdot \left(f_0 + \frac{\Delta f}{2} \right)} \quad \text{here, } f_2 < f_1, \quad f_0 = \frac{f_1 + f_2}{2} \quad \text{and} \quad \Delta f = f_1 - f_2 \\
 &= \frac{c \cdot \Delta f}{f_0^2 - \frac{1}{4} \Delta f^2} \stackrel{\Delta f \ll f_0}{\approx} \frac{c \cdot \Delta f}{f_0^2} \\
 &\approx \frac{\Delta f \cdot \lambda_0^2}{c}
 \end{aligned}$$

■ So, the spectral width $\Delta\lambda \approx \frac{\Delta f \cdot \lambda^2}{c} = \frac{10 \times 10^9 \times (1.55 \times 10^{-6})^2}{3 \times 10^8} \approx 0.08 \text{ nm}$

■ Broadening due to chromatic dispersion, $\Delta t_{CD} = 20 \frac{\text{ps}}{\text{km} \cdot \text{nm}} \times 0.08 \text{ nm} \times 0.625 \text{ km} = 1 \text{ ps}$

Problem 1: Dispersion

- In tutorial 3, we have seen that the impulse response of a Gaussian pulse after propagation through a fiber considering only chromatic dispersion can be written as,

$$|p_{\text{CD}}(t)| \sim \exp \left[-\frac{t^2}{2(T_0^2 + \Delta t_{\text{CD}}^2)} \right]$$

T_0 is initial pulse width

Δt_{CD} broadening due to chromatic dispersion

- In case we also considered modal dispersion the above equation would have the form

$$|p_{\text{CD}}(t)| \sim \exp \left[-\frac{t^2}{2(T_0^2 + \Delta t_{\text{CD}}^2 + \Delta t_{\text{MD}}^2)} \right]$$

- Total pulse broadening,

$$\Delta t_{\text{total}} = \sqrt{\Delta t_{\text{MD}}^2 + \Delta t_{\text{CD}}^2} = \sqrt{125^2 + 1^2} \approx 125.004 \text{ ps}$$

Problem 1: Dispersion

Broadening of 100 Gbit/s signal:

- In case of 100 Gbit/s the broadening due to modal dispersion remains unchanged.
- the spectral width of the signal changes to, $\Delta\lambda \approx 0.8 \text{ nm}$
- Pulse broadening due to chromatic dispersion, $\Delta t_{\text{CD}} = 10 \text{ ps}$
- Total pulse broadening, $\Delta t_{\text{total}} = \sqrt{\Delta t_{\text{MD}}^2 + \Delta t_{\text{CD}}^2} = \sqrt{125^2 + 10^2} \approx 125.4 \text{ ps}$
- From these examples, we see that in a multimode fiber, pulse broadening due to chromatic dispersion plays a role on the total broadening only at a high symbol rate.

Problem 2: Photodiode and Amplifiers

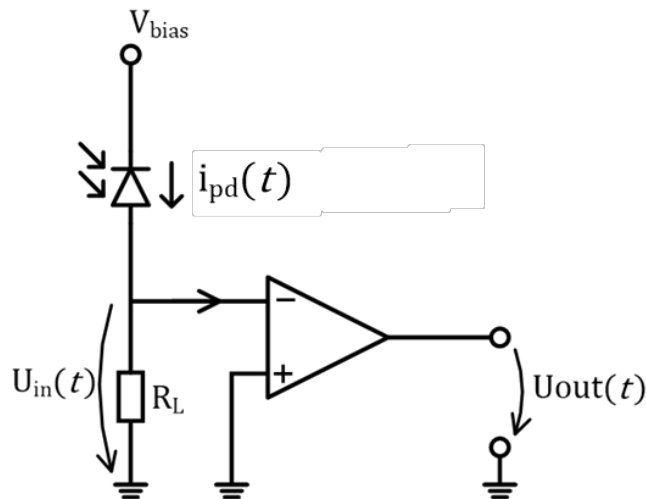
a) An InGaAs pin-photodiode has a responsivity of 0.8 A/W at a wavelength of 1.55 μm . If the optical power incident on the photodiode is -20 dBm, what is the output current?

■ The output current of a photodiode is to the input optical power as,

$$\begin{aligned} i &= R_d P_{\text{opt}} \\ &= 0.8 \frac{\text{A}}{\text{W}} \times 0.01 \text{mW} \\ &= 8 \mu\text{A} \end{aligned}$$

Problem 2: Photodiode and Amplifiers

b) Most often, photodetector outputs are further amplified by electrical amplification stages. Depict a high-impedance amplifier (HIA) setup. What is the main benefit of this amplifier setup? Given a load resistance $R_L = 10 \text{ k}\Omega$ and total capacitance of 1.6 pF (photodiode capacitance plus input capacitance of the amplifier), what is the 3-dB bandwidth of the HIA setup?



- At a finite temperature conductors will result in thermal noise
- Thermal noise variance, $\sigma_{th}^2 = \frac{4k_B T F_n \Delta f_n}{R_L}$
- Having large load resistance, R_L in HIA reduces thermal noise $\rightarrow$ makes the receiver more sensitive
- 3 dB bandwidth is given as:

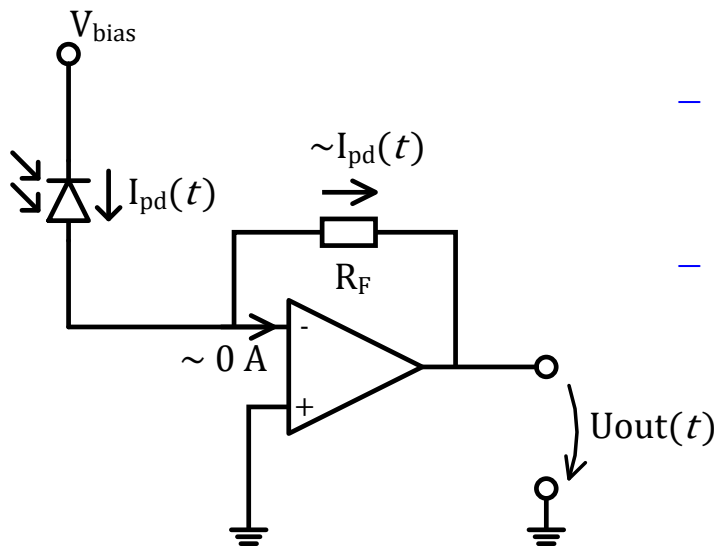
$$f_{3dB} = \frac{1}{2\pi R_L C}$$

$$= \frac{1}{2\pi \times 10 \times 10^3 \times 1.6 \times 10^{-12}}$$

$$\approx 10 \text{ MHz}$$

Problem 2: Photodiode and Amplifiers

- c) Depict a transimpedance amplifier (TIA) setup. What is the main benefit of this amplifier setup? Given an open loop gain $G = 100$, load resistance $R_F = 10 \text{ k}\Omega$ and total capacitance of 1.6 pF (photodiode capacitance plus input capacitance of the amplifier), what is the 3-dB bandwidth of the TIA setup?



- From the input side, the effect of the feedback resistance can be modelled by an equivalent load resistance of $R_L = R_F / (1 + G)$
- Due to the effective low resistance, they have a broader bandwidth than HIA. But TIA results in higher noise compared to HIA.
- 3 dB bandwidth is given as:

$$\begin{aligned}
 f_{3\text{dB}} &= \frac{1}{2\pi R_L C} \\
 &= \frac{1}{2\pi \times 99 \times 1.6 \times 10^{-12}} \\
 &\approx 1 \text{ GHz}
 \end{aligned}$$

Summary

- Effective differential group delay due to modal dispersion, $\Delta\tau_{\text{eff}} \sim L$
- In multi-mode fiber typically modal dispersion is more prominent than chromatic dispersion. Effect of chromatic dispersion scales in relation to the symbol rate.
- TIA provides higher bandwidth compared to HIA but results in more noise than HIA.

Optical Networks and Systems (ONS)

Tutorial 4

Tutorial Date: December 07, 2022, 14:00

Solution to Problem 1:

From the given figure, we can roughly estimate a 2 ns delay between the two modes for an 8.861 km fiber. The intermodal delay per km of propagation follows:

$$\Delta\tau_{\text{eff}} = \frac{2 \text{ ns}}{8.861 \text{ km}} = 0.23 \text{ ns/km}$$

Consider the case where a bit pattern in time is transmitted through a multi-mode fiber (MMF). We assume only the LP₀₁ and LP₁₁ modes are excited and both modes are excited by the same bit pattern and signal power. T is the symbol period $T = \frac{1}{\text{Bit Rate}} = 100 \text{ ps}$.

Now we further assume that the detection clock is synchronized to the LP₀₁ pulses. A detection error occurs when the two LP modes are delayed such that, the high level of a delayed mode is detected at the sampling instant of a zero level of the faster mode. Meaning, at the sampling instance t_2 , the pulse from LP₀₁ is differentially close to zero and the pulse from LP₁₁ is at a level of $\frac{A}{2}$.

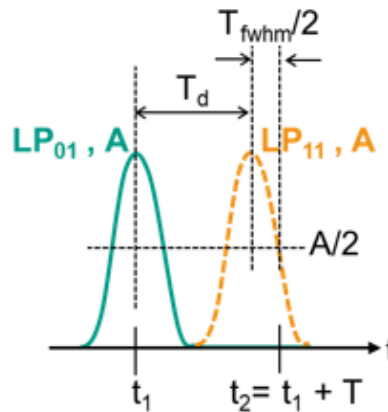


Figure 1

For the maximum tolerable delay T_d between the LP₀₁ and LP₀₂ follows:

$$T_d = T - \frac{T_{\text{fwhm}}}{2} = 100 \text{ ps} - 15 \text{ ps} = 85 \text{ ps}$$

With T_d determined and having 0.23 ns/km delay between the two modes, the modal dispersion limited reach at 10 Gbit/s can be found as

$$L = \frac{T_d}{\Delta\tau_{\text{eff}}} \approx 370 \text{ m}$$

Increasing the bit rate to 20 Gbit/s will reduce the length by a factor of 2 if the pulse width is halved as well.

Solution to Problem 2:

The RMS pulse width T_0 and FWHM pulse width T_{fwhm} is related as,

$$T_{\text{fwhm}} = 2T_0\sqrt{2\ln(2)}$$

From tutorial 2, we have seen that a Gaussian pulse in time domain can be written as,

$$p(t) = \exp\left(-4\ln(2)\frac{t^2}{T_{\text{fwhm}}^2}\right).$$

Substituting the FWHM pulse width with the RMS pulse width gives,

$$p(t) = \exp\left(-\frac{t^2}{2T_0^2}\right) \quad (1)$$

We used the following relation from a Fourier transform table to find the Fourier transform of a Gaussian pulse,

$$x(t) = \exp(-at^2) \xrightarrow{\mathcal{F}} X(f) = \sqrt{\frac{\pi}{a}} \exp\left(-\frac{\pi^2 f^2}{a}\right).$$

The Fourier transform of the Gaussian pulse $p(t)$ thus can be written as,

$$P(f) = \sqrt{2\pi}T_0 \exp\left(-2\pi^2 f^2 T_0^2\right). \quad (2)$$

Now, let us define the transmit signal. We consider a sequence $\{a_k\}_{k=-\infty}^{\infty}$ of transmit symbols spaced by a symbol interval $T_s = \frac{1}{R_s}$, where R_s is the symbol rate. The corresponding continuous time signal after pulse shaping can be written as,

$$s(t) = \sum_k a_k p(t - kT_s)$$

With an idealized electro-optic transmitter, we obtain the complex amplitude (electric field amplitude) of the light wave for a single polarization,

$$\underline{A}(z=0, t) = \underline{A}_0 \sum_k a_k p(t - kT_s)$$

Chromatic dispersion of a single-mode fiber of length L is described by a linear filter with the frequency response,

$$\underline{H}_{\text{CD}}(L, f) = \exp\left(\jmath \frac{1}{2} (2\pi f)^2 \beta_2 L\right)$$

We can apply the following Fourier transform relation to find the complex amplitude of the transmit signal in frequency domain,

$$\underline{A}(z, f) = \int_{-\infty}^{\infty} \underline{A}(z, t) \exp(-\jmath 2\pi f t) dt$$

After propagation through the optical fiber, the received complex amplitude in frequency domain can then be found as,

$$\begin{aligned} \underline{A}(z=L, f) &= \underline{A}(z=0, f) \cdot \underline{H}_{\text{CD}}(L, f) \\ &= \underline{A}_0 \cdot P(f) \cdot \underline{H}_{\text{CD}}(L, f) \cdot \sum_k a_k \exp(-\jmath 2\pi f k T_s) \end{aligned}$$

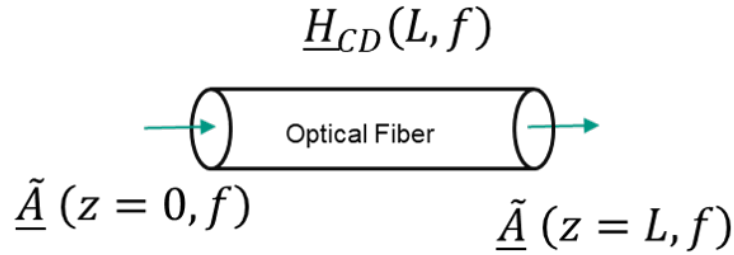


Figure 2: Linear propagation through an optical fiber

Here, $P(f)$ is the Fourier transform of $p(t)$. Combining the pulse shaping filter response and the effect of CD together to $\underline{P}_{CD}(f)$,

$$\begin{aligned}\underline{P}_{CD}(f) &= P(f) \cdot \underline{H}_{CD}(L, f) \\ &= \sqrt{2\pi}T_0 \exp\left(-2\pi^2 f^2 T_0^2\right) \cdot \exp\left(\alpha 2\pi^2 f^2 \beta_2 L\right) \\ &= \sqrt{2\pi}T_0 \exp\left[-2\pi^2 f^2 \left(T_0^2 - \alpha \beta_2 L\right)\right] \\ &= \frac{T_0}{T_L} \sqrt{2\pi}T_L \exp\left[-\frac{\pi^2 f^2}{\frac{1}{2T_L^2}}\right],\end{aligned}$$

here, $T_L = \sqrt{T_0^2 - \alpha \beta_2 L}$ Inverse fourier transform then gives the impulse response,

$$\underline{p}_{CD}(t) = \frac{T_0}{T_L} \exp\left[-\frac{t^2}{2T_L^2}\right] \quad (3)$$

We consider the absolute value of the impulse response to find the broadening of the pulse duration and obtain,

$$\begin{aligned}|\underline{p}_{CD}(t)| &= \left|\frac{T_0}{T_L}\right| \cdot \left|\exp\left[-\frac{t^2}{2T_L^2}\right]\right| \\ &= \left|\frac{T_0}{T_L}\right| \exp\left[-\frac{t^2}{2} \Re\left(\frac{1}{T_L^2}\right)\right]\end{aligned}$$

$$\begin{aligned}\left|\frac{T_0}{T_L}\right| &= \left|\frac{T_0}{\sqrt{T_0^2 - \alpha \beta_2 L}}\right| \\ &= \left|\frac{T_0}{T_0 \sqrt{1 - \frac{\alpha \beta_2 L}{T_0^2}}}\right| \\ &= \frac{1}{\sqrt[4]{1 + \frac{\beta_2^2 L^2}{T_0^4}}}\end{aligned}$$

$$\begin{aligned}\Re\left(\frac{1}{T_L^2}\right) &= \Re\left(\frac{1}{T_0^2 - \alpha\beta_2 L}\right) \\ &= \Re\left(\frac{T_0^2 + \alpha\beta_2 L}{T_0^4 + \beta_2^2 L^2}\right) \\ &= \frac{1}{T_0^2 \left(1 + \frac{\beta_2^2 L^2}{T_0^4}\right)}\end{aligned}$$

Finally substituting these values we can write,

$$\left|p_{CD}(t)\right| = \frac{1}{\sqrt[4]{1 + \frac{\beta_2^2 L^2}{T_0^4}}} \exp\left[-\frac{t^2}{2T_0^2 \left(1 + \frac{\beta_2^2 L^2}{T_0^4}\right)}\right] \quad (4)$$

Let us represent, $\eta_{CD}^2 = 1 + \frac{\beta_2^2 L^2}{T_0^4}$ and replace the RMS width with the FWHM pulse width,

$$\begin{aligned}\eta_{CD} &= \sqrt{1 + \frac{\beta_2^2 L^2}{T_0^4}} \\ &= \sqrt{1 + \frac{(8 \ln(2)\beta_2 L)^2}{T_{fwhm}^4}}\end{aligned}$$

Using η_{CD} , the absolute value of $p_{CD}(t)$ can be further written as,

$$\left|p_{CD}(t)\right| = \frac{1}{\sqrt{\eta_{CD}}} \cdot \exp\left[-4 \ln(2) \frac{t^2}{(\eta_{CD} T_{fwhm})^2}\right] \quad (5)$$

So we see that the received signal also has a gaussian pulse shape whose pulse width is broadened by a factor of η_{CD} . To receive the signal without any error (without ISI), the received pulse width must not be larger than double the symbol interval, i.e., $\eta_{CD} T_{fwhm} \leq 2T_s$. This can be seen in the sketch of 3 and can be written as,

$$\begin{aligned}\sqrt{1 + \frac{(8 \ln(2)\beta_2 L)^2}{T_{fwhm}^4}} \cdot T_{fwhm} &\leq 2T_s \\ L &\leq \frac{T_{fwhm}^2}{8 \ln(2)\beta_2} \sqrt{\left[\left(\frac{2T_s}{T_{fwhm}}\right)^2 - 1\right]} \\ &\leq \frac{2\pi c_0 \gamma \sqrt{4 - \gamma^2}}{8 \ln(2)\lambda^2 D_{CD} R_s^2} \\ &\leq \frac{\pi c_0 \gamma \sqrt{4 - \gamma^2}}{4 \ln(2)\lambda^2 D_{CD} R_s^2} \\ L_{CD} &= \frac{\pi c_0 \gamma \sqrt{4 - \gamma^2}}{4 \ln(2)\lambda^2 D_{CD} R_s^2}\end{aligned}$$

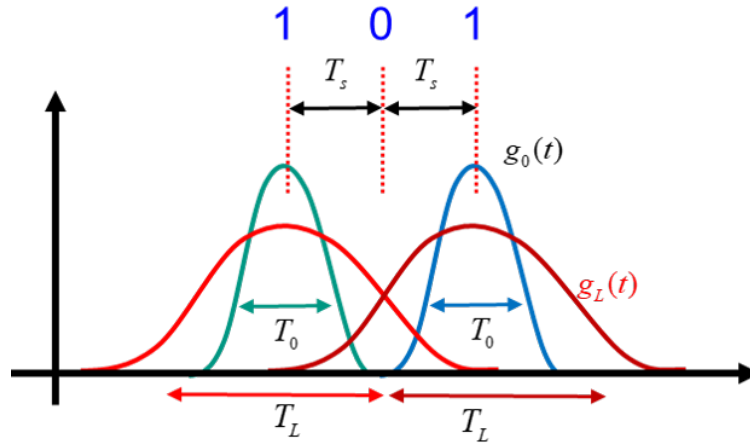


Figure 3: Sketch showing the effect of broadening of a pulse.

Here, L_{CD} is the maximum distance for which error free transmission can be achieved. The parameters β_2 and γ is defined as $\beta_2 = \frac{\lambda^2 D_{CD}}{2\pi c_0}$ and $\gamma = \frac{T_{fwhm}}{T_s}$. The maximum reach for the example parameters given in the problem set is presented in the following table,

R_s	10 Gbit/s	20 Gbit/s
T_0	30 ps	15 ps
T_{fwhm}	70 ps	35 ps
T_s	100 ps	50 ps
L_{CD}	93.51 km	23.37 km